

Investing Money - Simple Interest

Simple Interest is called simple interest because at the end of each year no interest is paid, it is only paid when the total time period of the investment is reached.

$$I = Prn$$

There are 4 variables in the formula, these are:

I = Interest earned on the investment (or payable on a loan)

P = Principal, the amount invested (or borrowed if a loan)

r = interest rate applied to the investment (or loan), expressed as a decimal (not a percentage)

n = the term, which is the time period the investment (or loan) is over

To find the interest payable, multiply the Principal, rate (as a decimal) and term. There is an example below. As you are dealing with money remember to round your answer to 2 decimal places (2 d.p.)

Example: Find the interest earned when \$5 100 is invested
for 6 years at 7% p.a.

$$r = 7 \div 100 = 0.07$$

Convert 7% to a decimal by $\div 100$

$$I = Prn$$

Write the formula

$$= 5\,100 \times 0.07 \times 6$$

Substitute values into formula

$$I = \$2\,142$$

Answer with '\$' sign and cents to
2 decimal places (if needed)

The second step, if required is finding the account balance (A). Which is the amount you have in the account after the interest is paid. The account balance is found by adding the principal and the interest together. So:

$$A = P + I$$

In the example above. The account balance would be:

$$A = P + I = 5\,100 + 2\,142 = \$7\,242$$

Investing Money - Simple Interest (Sheet 1)

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Convert these interest rates from percentages to decimals. Divide by 100 to get a decimal.

Example:

$$16\% = 16 \div 100 = 0.16$$

1 $9\% = \boxed{} \div 100 = \boxed{}$

2 $5\% = \boxed{} \div \boxed{} = \boxed{}$

3 $10\% = \boxed{} \div \boxed{} = \boxed{}$

Answer only, from now on.

4 $14.5\% = \boxed{}$

5 $12.7\% = \boxed{}$

6 $4.92\% = \boxed{}$

7 $0.5\% = \boxed{}$

8 $7\frac{1}{4}\% = \boxed{}$

Calc:

SHIFT $\frac{1}{x}$ 7 $\rightarrow$ 1 $\rightarrow$ 4 $\rightarrow$ $\div$ 1 0 0
S $\leftrightarrow$ D $\div$

9 $6\frac{3}{4}\% = \boxed{}$

Simple interest is found using $I = Prn$, but first convert the rate R to a decimal. Find I .

Example: Find the interest earned:

$$P = \$950, R = 8\% \text{ p.a.}, n = 5 \text{ years}$$

$$r = 8 \div 100 = 0.08$$

$$I = Prn$$

$$= 950 \times 0.08 \times 5$$

$$I = \$380$$

$$P = \$8\,300, R = 6\% \text{ p.a.}, n = 4 \text{ years}$$

10 $r = \boxed{} \div 100 = \boxed{}$

$$I = Prn$$

$$= \boxed{} \times \boxed{} \times \boxed{}$$

P

r

n

$$I = \boxed{}$$

Show the '\$' sign

$$P = \$970, R = 4.5\% \text{ p.a.}, n = 2 \text{ years}$$

11 $r = \boxed{} \div \boxed{} = \boxed{}$

$$I = Prn$$

$$= \boxed{} \times \boxed{} \times \boxed{}$$

$I = \boxed{}$ Cents are always 2 d.p.

$$P = \$5\,400, R = 6.04\% \text{ p.a.}, n = 7 \text{ yrs}$$

12 $r = \boxed{}$

$$I = Prn$$

$$= \boxed{}$$

$$I = \boxed{}$$

The next three have smaller interest rates, but you still have to $\div 100$, as they are percentages

$$P = \$820, R = 0.15\%/\text{week}, n = 9 \text{ wks}$$

13 $r = \boxed{}$

$$I = Prn$$

$$= \boxed{}$$

$$I = \boxed{}$$

$$P = \$11\,600, R = 0.4\%/\text{mth}, n = 8 \text{ mths}$$

14 $\boxed{}$

Convert R to r (% to decimal)

Write Simple Interest Formula

Make the Substitution

Show ' I ', the '\$' sign and cents to 2 d.p.

$$P = \$450, R = 0.05\%/\text{wk}, n = 19 \text{ weeks}$$

15 $\boxed{}$

Round to 2 d.p.
NOT nearest 5c.

$$P = \$47\,130, R = 8.4\% \text{ p.a.}, n = 6 \text{ years}$$

16 $\boxed{}$

Keep going, but do less work. Use 3 lines not 4, Convert R to r , in the substitution line.

$$P = \$1\,720, R = 4\% \text{ p.a.}, n = 5 \text{ yrs}$$

17 $I = Prn$ Use r not R

$$= \boxed{} \times \boxed{} \times \boxed{}$$

$$I = \boxed{}$$

$$P = \$9\,050, R = 3.75\% \text{ p.a.}, n = 4 \text{ yrs}$$

18 $I = Prn$ Use r not R

$$= \boxed{} \times \boxed{} \times \boxed{}$$

$$I = \boxed{}$$

$$P = \$180, R = 6.25\% \text{ p.a.}, n = 4.5 \text{ yrs}$$

19 $\boxed{}$

$$P = \$57\,000, R = 4.02\% \text{ p.a.}, n = 9 \text{ yrs}$$

20 $\boxed{}$

$$P = \$685.12, R = 13.07\% \text{ p.a.}, n = 3 \text{ yrs}$$

21 $\boxed{}$

Investing Money - Simple Interest - Account Balance (Sheet 2)

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The account balance 'A' is what is in your account after the interest is paid. Calculate A.

Example: $P = \$1\,300$, $I = \$275.20$

$$A = P + I = 1\,300 + 275.20$$

$$A = \$1\,575.20$$

$P = \$1\,150$, $I = \$662.30$

1 $A = \boxed{} + \boxed{}$

$$A = \boxed{}$$

Show '\$' sign and cents to 2 decimal places

$P = \$580$, $I = \$39.65$

2 $A = \boxed{} + \boxed{}$

$$A = \boxed{}$$

$P = \$56\,970$, $I = \$4\,076.10$

3 $A = \boxed{}$

$$A = \boxed{}$$

$P = \$8\,030.62$, $I = \$1\,510.18$

4 $\boxed{}$

$P = \$3\,877$, $I = \$772.92$

5 $\boxed{}$

Jen has \$458.08 in her bank account and receives \$14.32 interest. Find her new account balance.

6 $\boxed{}$

Neil has \$2 504.24 in the bank. He then receives interest of \$83.19. Calculate his account balance.

7 $\boxed{}$

Calculate the interest earned then add it to the principal to find the account balance.



Example:

$P = \$1\,400$, $R = 7\%$ p.a., $n = 5$ years

$$I = Prn$$

$$= 1\,400 \times 0.07 \times 5$$

$$I = \$490$$

$$A = 1\,400 + 490 = \$1\,890$$

$P = \$4\,200$, $R = 5.2\%$ p.a., $n = 3$ yrs

8 $I = Prn$ *Use r not R*

$$= \boxed{} \times \boxed{} \times \boxed{}$$

$$I = \boxed{}$$

Show '\$' sign and cents to 2 decimal places

$$A = \boxed{} + \boxed{}$$

$$A = \boxed{}$$

$P = \$19\,050$, $R = 4.8\%$ p.a., $n = 4$ yrs

9 $I = Prn$

$$= \boxed{} \times \boxed{} \times \boxed{}$$

$$I = \boxed{}$$

$$A = \boxed{} + \boxed{}$$

$$A = \boxed{}$$

$P = \$1\,150$, $R = 5.25\%$ p.a., $n = 6$ yrs

10 $\boxed{}$

Interest rates are smaller as the period is shorter. Find the interest and account balance.



$P = \$79\,500$

$R = 0.05\%$ per week $n = 38$ weeks

Remember to convert R to r (% to a decimal)

11 $\boxed{}$

$P = \$10\,200$

$R = 0.7\%$ per month $n = 9$ months

12 $\boxed{}$

Jason invests \$840 in an account for 7 months which pays 0.45% per month. Calculate:

13 The interest Jason has earned

14 His account balance once the interest has been credited.

Investing Money - Simple Interest (Sheet 1)

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Convert these interest rates from percentages to decimals. Divide by 100 to get a decimal.

Example:

$$16\% = 16 \div 100 = 0.16$$

1 $9\% = \boxed{9} \div 100 = \boxed{0.09}$

2 $5\% = \boxed{5} \div 100 = \boxed{0.05}$

3 $10\% = \boxed{10} \div 100 = \boxed{0.1}$

Answer only, from now on.

4 $14.5\% = \boxed{0.145}$

5 $12.7\% = \boxed{0.127}$

6 $4.92\% = \boxed{0.0492}$

7 $0.5\% = \boxed{0.005}$

8 $7\frac{1}{4}\% = \boxed{0.0725}$

Calc:

SHIFT $\frac{1}{x}$ 7 $\frac{1}{x}$ 1 $\frac{1}{x}$ 4 $\frac{1}{x}$ $\div$ 1 0 0

S $\leftrightarrow$ D $\frac{1}{x}$

9 $6\frac{3}{4}\% = \boxed{0.0675}$

Simple interest is found using $I = Prn$, but first convert the rate R to a decimal. Find I .

Example: Find the interest earned:

$$P = \$950, R = 8\% \text{ p.a.}, n = 5 \text{ years}$$

$$r = 8 \div 100 = 0.08$$

$$I = Prn$$

$$= 950 \times 0.08 \times 5$$

$$I = \$380$$

$$P = \$8\,300, R = 6\% \text{ p.a.}, n = 4 \text{ years}$$

10 $r = \boxed{6} \div 100 = \boxed{0.06}$

$$I = Prn$$

$$= \boxed{8\,300} \times \boxed{0.06} \times \boxed{4}$$

P

r

n

$$I = \boxed{\$1\,992}$$

Show the '\$' sign

$$P = \$970, R = 4.5\% \text{ p.a.}, n = 2 \text{ years}$$

11 $r = \boxed{4.5} \div \boxed{100} = \boxed{0.045}$

$$I = Prn$$

$$= \boxed{970} \times \boxed{0.045} \times \boxed{2}$$

$$I = \boxed{\$87.30}$$

Cents are always 2 d.p.

$$P = \$5\,400, R = 6.04\% \text{ p.a.}, n = 7 \text{ yrs}$$

12 $r = \boxed{6.04 \div 100 = 0.0604}$

$$I = Prn$$

$$= \boxed{5\,400} \times \boxed{0.0604} \times \boxed{7}$$

$$I = \boxed{\$2\,283.12}$$

The next three have smaller interest rates, but you still have to $\div 100$, as they are percentages

$$P = \$820, R = 0.15\%/\text{week}, n = 9 \text{ wks}$$

13 $r = \boxed{0.15 \div 100 = 0.0015}$

$$I = Prn$$

$$= \boxed{820} \times \boxed{0.0015} \times \boxed{9}$$

$$I = \boxed{\$11.07}$$

$$P = \$11\,600, R = 0.4\%/\text{mth}, n = 8 \text{ mths}$$

14 $r = \boxed{0.4 \div 100 = 0.004}$

Convert R to r (% to decimal)

$$I = Prn$$

Write Simple Interest Formula

$$= \boxed{11\,600} \times \boxed{0.004} \times \boxed{8}$$

Make the Substitution

$$I = \$371.20$$

Show 'I=', the '\$' sign and cents to 2 d.p.

$$P = \$450, R = 0.05\%/\text{wk}, n = 19 \text{ weeks}$$

15 $r = \boxed{0.05 \div 100 = 0.0005}$

$$I = Prn$$

$$= \boxed{450} \times \boxed{0.0005} \times \boxed{19}$$

$$I = \$4.28$$

Round to 2 d.p.
NOT nearest 5c.

$$P = \$47\,130, R = 8.4\% \text{ p.a.}, n = 6 \text{ years}$$

16 $r = \boxed{8.4 \div 100 = 0.084}$

$$I = Prn$$

$$= \boxed{47\,130} \times \boxed{0.084} \times \boxed{6}$$

$$I = \$23\,753.52$$

Keep going, but do less work. Use 3 lines not 4, Convert R to r , in the substitution line.

$$P = \$1\,720, R = 4\% \text{ p.a.}, n = 5 \text{ yrs}$$

17 $I = Prn$ Use r not R

$$= \boxed{1\,720} \times \boxed{0.04} \times \boxed{5}$$

$$I = \$344$$

$$P = \$9\,050, R = 3.75\% \text{ p.a.}, n = 4 \text{ yrs}$$

18 $I = Prn$ Use r not R

$$= \boxed{9\,050} \times \boxed{0.0375} \times \boxed{4}$$

$$I = \$1\,357.50$$

$$P = \$180, R = 6.25\% \text{ p.a.}, n = 4.5 \text{ yrs}$$

19 $I = Prn$

$$= \boxed{180} \times \boxed{0.0625} \times \boxed{4.5}$$

$$I = \$50.63$$

$$P = \$57\,000, R = 4.02\% \text{ p.a.}, n = 9 \text{ yrs}$$

20 $I = Prn$

$$= \boxed{57\,000} \times \boxed{0.0402} \times \boxed{9}$$

$$I = \$20\,622.60$$

$$P = \$685.12, R = 13.07\% \text{ p.a.}, n = 3 \text{ yrs}$$

21 $I = Prn$

$$= \boxed{685.12} \times \boxed{0.1307} \times \boxed{3}$$

$$I = \$268.64$$

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The account balance 'A' is what is in your account after the interest is paid. Calculate A.

Example: $P = \$1\,300$, $I = \$275.20$

$$A = P + I = 1\,300 + 275.20$$

$$A = \$1\,575.20$$

$P = \$1\,150$, $I = \$662.30$

1 $A = \boxed{1\,150} + \boxed{662.30}$

$$A = \boxed{\$1\,812.30}$$

Show '\$' sign and cents to 2 decimal places

$P = \$580$, $I = \$39.65$

2 $A = \boxed{580} + \boxed{39.65}$

$$A = \boxed{\$619.65}$$

$P = \$56\,970$, $I = \$4\,076.10$

3 $A = 56\,970 + 4\,076.10$

$$A = \$61\,046.10$$

$P = \$8\,030.62$, $I = \$1\,510.18$

4 $A = 8\,030.62 + 1\,510.18$

$$A = \$9\,540.80$$

$P = \$3\,877$, $I = \$772.92$

5 $A = 3\,877 + 772.92$

$$A = \$4\,649.92$$

Jen has \$458.08 in her bank account and receives \$14.32 interest. Find her new account balance.

6 $A = 458.08 + 14.32$

$$A = \$472.40$$

Neil has \$2 504.24 in the bank. He then receives interest of \$83.19. Calculate his account balance.

7 $A = 2\,504.24 + 83.19$

$$A = \$2\,587.43$$

Calculate the interest earned then add it to the principal to find the account balance.



Example:

$P = \$1\,400$, $R = 7\%$ p.a., $n = 5$ years

$$I = Prn$$

$$= 1\,400 \times 0.07 \times 5$$

$$I = \$490$$

$$A = 1\,400 + 490 = \$1\,890$$

$P = \$4\,200$, $R = 5.2\%$ p.a., $n = 3$ yrs

8 $I = Prn$ *Use r not R*

$$= \boxed{4\,200} \times \boxed{0.052} \times \boxed{3}$$

$$I = \boxed{\$655.20}$$

Show '\$' sign and cents to 2 decimal places

$$A = \boxed{4\,200} + \boxed{655.20}$$

$$A = \boxed{\$4\,855.20}$$

$P = \$19\,050$, $R = 4.8\%$ p.a., $n = 4$ yrs

9 $I = Prn$

$$= \boxed{19\,050} \times \boxed{0.048} \times \boxed{4}$$

$$I = \boxed{\$3\,657.60}$$

$$A = \boxed{19\,050} + \boxed{3\,657.60}$$

$$A = \boxed{\$22\,707.60}$$

$P = \$1\,150$, $R = 5.25\%$ p.a., $n = 6$ yrs

10 $I = Prn$

$$= 1\,150 \times 0.0525 \times 6$$

$$I = \$362.25$$

$$A = 1\,150 + 362.25$$

$$A = \$1\,512.25$$

Interest rates are smaller as the period is shorter. Find the interest and account balance.



$P = \$79\,500$

$R = 0.05\%$ per week $n = 38$ weeks

Remember to convert R to r (% to a decimal)

11 $I = Prn$

$$= 79\,500 \times 0.0005 \times 38$$

$$I = \$1\,510.50$$

$$A = 79\,500 + 1\,510.50$$

$$A = \$81\,010.50$$

$P = \$10\,200$

$R = 0.7\%$ per month $n = 9$ months

12 $I = Prn$

$$= 10\,200 \times 0.007 \times 9$$

$$I = \$642.60$$

$$A = 10\,200 + 642.60$$

$$A = \$10\,842.60$$

Jason invests \$840 in an account for 7 months which pays 0.45% per month. Calculate:

13 The interest Jason has earned

$$I = Prn$$

$$= 840 \times 0.0045 \times 7$$

$$I = \$26.46$$

14 His account balance once the interest has been credited.

$$A = 840 + 26.46$$

$$A = \$866.46$$